Ans.1 Differential Equation:

$$\boxed{y''' + 3y'' + 5y' + y = x''' + 4x'' + 6x' + 8x}$$ Ans.

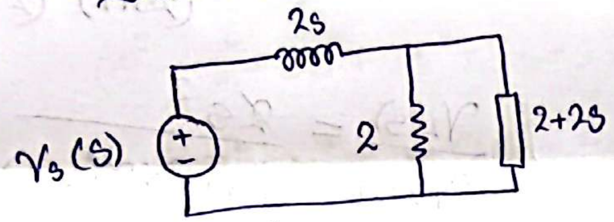
In S-Domain (transfer function):

$$s^3 Y(s) + 3s^2 Y(s) + 5s Y(s) + Y(s) = s^3 X(s) + 4s^2 X(s) + 6s X(s) + 8X(s)$$

$$(s^3 + 3s^2 + 5s + 1) Y(s) = (s^3 + 4s^2 + 6s + 8) X(s)$$

$$\boxed{\frac{Y(s)}{X(s)} = \frac{s^3 + 4s^2 + 6s + 8}{s^3 + 3s^2 + 5s + 1}}$$ Ans.

Ans.2 1 First in S-Domain:



$$\frac{1}{Z_A} = \frac{1}{2} + \frac{1}{2+2s}$$

$$Z_A = \frac{2 \times (2+2s)}{2+2s+2}$$

$$Z_A = \frac{\cancel{2}(2+2s)}{\cancel{2}(s+2)}$$

$$\boxed{Z_A = \frac{2s+2}{s+2}}$$

$$V_A(s) = V_s(s) \times \frac{\frac{2s+2}{s+2}}{2s + \frac{2s+2}{s+2}}$$

$$V_A(s) = V_s(s) \times \frac{\frac{2s+2}{\cancel{s+2}}}{\frac{2s^2+4s+2s+2}{\cancel{s+2}}}$$

$$V_A(s) = V_s(s) \times \frac{\cancel{2}(s+1)}{\cancel{2}(s^2+3s+1)}$$

$$\boxed{V_A(s) = V_s(s) \times \frac{(s+1)}{(s^2+3s+1)}}$$

$$V_L(s) = V_A(s) \times \frac{2s}{2s+2}$$

$$V_L(s) = V_s(s) \times \frac{\cancel{s+1}}{s^2+3s+1} \times \frac{\cancel{2}s}{\cancel{2}(\cancel{s+1})}$$

$$\boxed{\frac{V_L(s)}{V_s(s)} = \frac{s}{s^2+3s+1}}$$ Ans.

Now Differential Equation:

$$\boxed{v_L''(t) + 3v_L'(t) + v_L(t) = v_s'(t)}$$ Ans.

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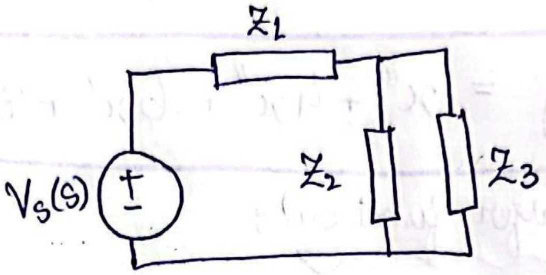
② First in S-Domain:

$Z_1 = 2 + \frac{2}{s}$

$Z_2 = 2 + \frac{2}{s}$

$Z_3 = 2 + 2s$



$$Z_2 \parallel Z_3 = \frac{1}{2+\frac{2}{s}} + \frac{1}{2+2s}$$

$$\frac{1}{Z_A} = \frac{2+2s+(2s+2)/s}{(2s+2)/s \,*\, (2+2s)}$$

$$\frac{1}{Z_A} = \frac{\frac{2s+2s^2+2s+2}{\cancel{s}}}{\frac{(2s+2)\times(2+2s)}{\cancel{s}}}$$

$$\frac{1}{Z_A} = \frac{2(s^2+2s+1)}{4s+4+4s^2+4s}$$

$$\frac{1}{Z_A} = \frac{\cancel{2}\,\cancel{(s^2+2s+1)}}{\cancel{4}_{2}\,\cancel{(s^2+2s+1)}}$$

$$\boxed{Z_A = 2}$$

$$V_A(s) = V_s(s) \times \frac{2}{2+2+\frac{2}{s}}$$

$$V_A(s) = V_s(s) \times \frac{2}{\frac{2s+2s+2}{s}}$$

$$V_A(s) = V_s(s) \times \frac{\cancel{2}}{\frac{(2s+1)\cancel{2}}{s}}$$

$$\boxed{V_A(s) = V_s(s) \times \frac{s}{(2s+1)}}$$

$$V_L(s) = V_A(s) \times \frac{2s}{2+2s}$$

$$V_L(s) = V_s(s) \times \frac{s}{(2s+1)} \times \frac{2s}{(2+2s)}$$

$$\frac{V_L(s)}{V_s(s)} = \frac{\cancel{2}s^2}{(2s+1)\times\cancel{2}(s+1)}$$

$$\frac{V_L(s)}{V_s(s)} = \frac{s^2}{2s^2+s+2s+1}$$

$$\boxed{\frac{V_L(s)}{V_s(s)} = \frac{s^2}{2s^2+3s+1}}$$ Ans.

Now Differential Equation:

$$\boxed{2v_L''(t) + 3v_L'(t) + v_L(t) = v_s''(t)}$$

Ans.

in S-Domain:

$$F(s) = 5X(s) + 5s^2 X(s) + 4s\, X(s)$$

$$F(s) = (5 + 5s^2 + 4s)\, X(s)$$

$$\boxed{\frac{X(s)}{F(s)} = \frac{1}{5s^2+4s+5}}$$ Ans.

Now Differential Equation:

$$\boxed{f(t) = 5x''(t) + 4x'(t) + 5x(t)}$$ Ans.

Ans. 3

$$\frac{C(s)}{R(s)} = \frac{s^4+3s^3+2s^2+s+1}{s^5+4s^4+3s^3+2s^2+3s+2}$$

$$(s^5+4s^4+3s^3+2s^2+3s+2)\, C(s) = (s^4+3s^3+2s^2+s+1)\, R(s)$$

Converting to a Differential-Equation:

$$c'''''(t) + 4c''''(t) + 3c'''(t) + 2c''(t) + 3c'(t) + 2c(t) =$$
$$r''''(t) + 3r'''(t) + 2r''(t) + r'(t) + r(t)$$

$\Rightarrow$ Input $r(t) = 3t^3$

$$r(t) = 3t^3$$
$$r'(t) = 9t^2$$
$$r''(t) = 18t$$
$$r'''(t) = 18$$
$$r''''(t) = 0$$

$\therefore$
$$\boxed{\begin{gathered} c'''''(t) + 4c''''(t) + 3c'''(t) + 2c''(t) + 3c'(t) + 2c(t) = \\ 3t^3 + 9t^2 + 36t + 54 \end{gathered}}$$ Ans.

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